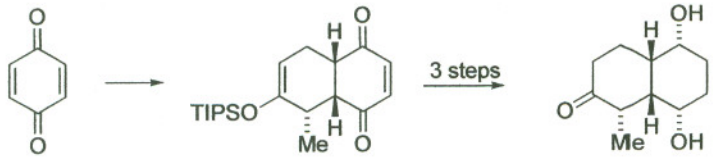
Organic Cume
Oct 29, 2004

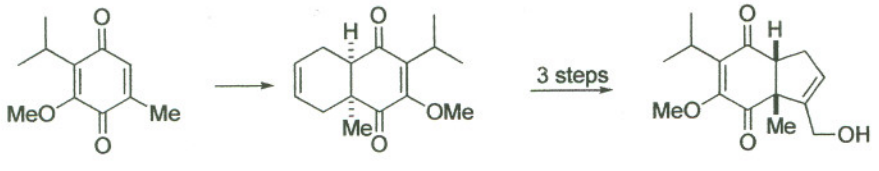
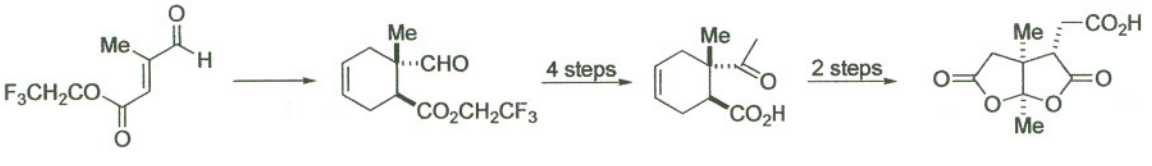
The Diels-Alder reaction has been used in many hundreds of natural product syntheses. In a recent paper, E J Corey showed how asymmetric catalysis in the key step of classical synthesis of racemic products transforms these classic syntheses into enantioselective ones (*JACS* **2004**, *126,* 13708). Use your knowledge of organic chemistry to fill in as many synthetic and mechanistic details as you can, for each of the following transformations, taken from some of these "classic" syntheses. You do not need to consider absolute configuration, but you do need to consider relative configuration and regioselectivity.

The first step in each sequence is a Diels-Alder reaction. Show the correct diene, explain the relative configuration in the products, and then suggest ways to do the next transformations. Be sure to consider functional group compatibility in your synthetic sequences.

1. (25 pts) Sarrett synthesis of cortisone



2. (25 pts) Kende synthesis of dendrobine



3. (50 pts) Eschenmoser's synthesis of Vitamin B_{12}.